

LastMile

Delivery Layer for Humanitarian Cash Assistance

Project Proposal

Prepared for SAPCONE - Turkana County, Kenya

Group 2 Members

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Problem Statement

SAPCONE faces severe operational and security challenges in delivering cash assistance to digitally excluded participants in remote Turkana County, who lack the smartphones and network connectivity required for direct digital disbursements. To circumvent this, the organization relies on vulnerable manual workarounds: field staff must transport large sums of physical cash to remote locations, or participants must nominate proxies with basic mobile phones to receive and deliver funds on their behalf. Compounding this issue, relying on physical cards or paper vouchers for identification is highly impractical for nomadic pastoralist communities, as these items are easily lost, damaged, or expensive to issue at scale.

These proxy-based and manual distribution methods create a critical deficit in accountability and verifiability. Existing tracking tools only log transfers after the fact rather than gating payments on actual proof of physical handover, providing donors with limited assurance that the cash successfully reached the intended recipient. Consequently, SAPCONE struggles to verify final-mile proxy deliveries while simultaneously trying to uphold strict privacy requirements that isolate participant personally identifiable information (PII) from public and operational transaction records.

The Solution

To address the challenges of reaching digitally excluded communities in Turkana County, we propose the implementation of LastMile, a secure, offline-first, last-mile delivery layer. This system acts as a human-digital bridge, combining offline field tools, SMS/USSD gateways, and blockchain technology to guarantee secure, verifiable, and privacy-preserving cash handovers.

The solution removes complex field cryptography in favor of a Trust-Based Proxy Model, built on five core pillars:

- **Universal Offline-First Registration:** Using a mobile application, field staff can register participants completely offline. The app captures essential data (Participant Name, Reference ID, Currency preference, Phone Number) and a critical Proxy Checkbox. The device queues this data and automatically syncs with the central backend once internet connectivity is restored.
- **Trusted Proxy Communications Channel:** The system embraces the reality that participants and proxies operate on mutual trust, ensuring the last-mile handover is as frictionless as possible. For proxy-dependent participants, the proxy's feature phone acts as the sole communications channel. When funds are ready, an SMS notification containing the transaction One-Time Password (OTP) is sent directly to the proxy.
- **Universal USSD Wallet Access:** Participants and proxies interact with the system via a simple USSD menu (e.g., *340#). The user inputs the Reference ID and the received OTP to authenticate the session, allowing them to access funds or check balances on basic feature phones without needing a smartphone app.
- **Automated Fiat Off-Ramping:** Once a transaction is authorized via USSD using the valid OTP, the backend instantly routes the digital aid (USDC) through a mobile money aggregator. The funds are automatically converted to the participant's registered local currency (e.g., KES) and deposited into the active phone number for physical cash withdrawal.
- **Human-Gated Clawbacks & Auditability:** The SAPCONE Admin Portal provides real-time monitoring of all disbursements. If a payout remains unclaimed, the system automatically sends SMS reminders. However, funds are never automatically returned. An administrator must manually review flagged stagnant accounts and authorize a clawback withdrawal, ensuring strict human oversight.

Stakeholders

Stakeholder	Responsibilities
Admin	Creates programmes; oversees registrations and deliveries; explicitly reviews and authorizes clawbacks for uncollected funds; generates reports; audits completed distributions.
Registrar / Financial Officer	Registers participants; synchronizes offline registrations; issues Reference IDs.
Proxy	Acts as the communications channel; receives delivery SMS alerts and OTPs; facilitates USSD access; delivers cash to the participant based on mutual trust.
Participant	Partners with the proxy (where applicable) to access funds; receives physical cash.

Registration Data Model

Every participant is registered through the same unified form, regardless of whether they will access funds directly or through a proxy. The fields captured are:

Field	Purpose
Participant Name	Personal identity capture. Stored separately from transactional records (PII isolation).
Reference ID	System-generated unique identifier used for all transactional/operational references. No PII on this record.
Currency	Local currency the disbursement should be converted to during off-ramp (e.g., KES).
Phone Number	The number funds/notifications are sent to. Can be the participant's own number, or a proxy's number.
Proxy Checkbox	Marks whether the phone number above belongs to a proxy rather than the participant. Determines which of the two access flows below applies.

System Architecture

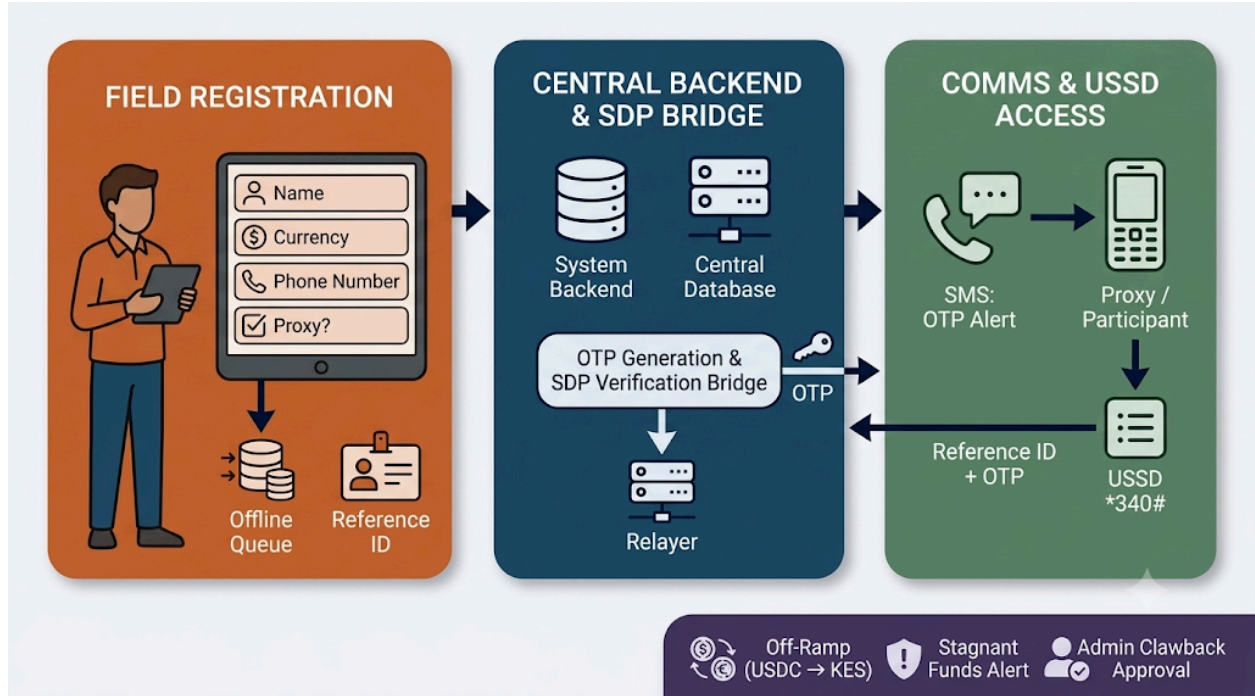


Figure 1 - LastMile system architecture.

User & Delivery Flows

The following flowchart maps the participant-centered delivery and verification workflow, from field registration through on-chain finalization.

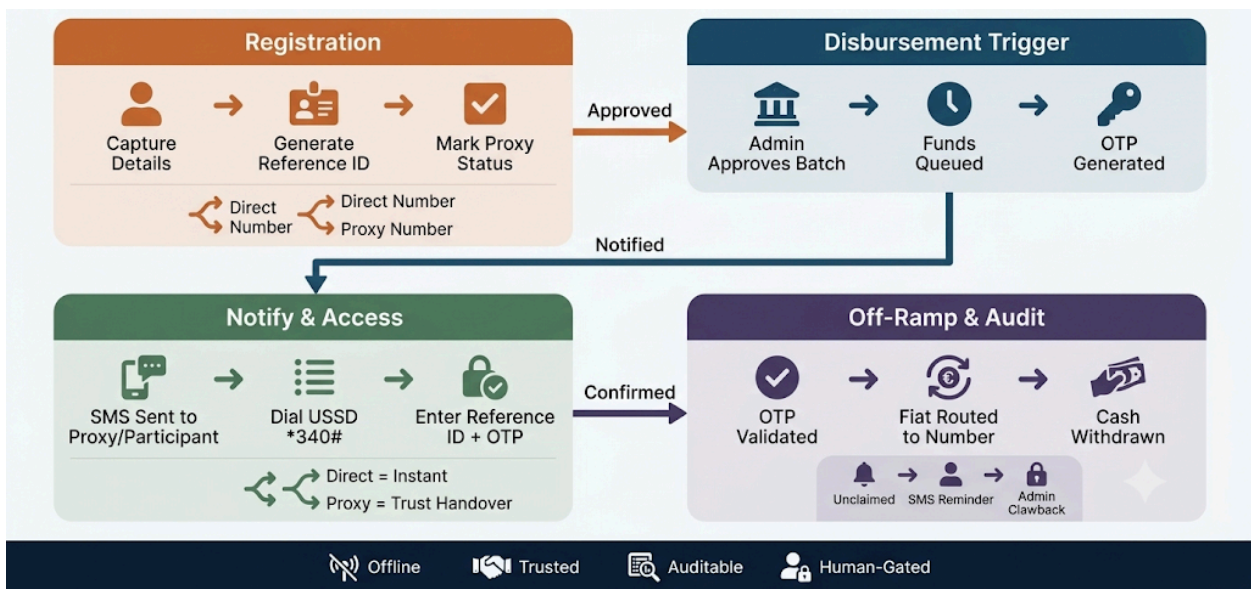


Figure 2 - LastMile humanitarian aid user flow.

The figure below summarizes the same journey by actor, channel, and system component at each step.

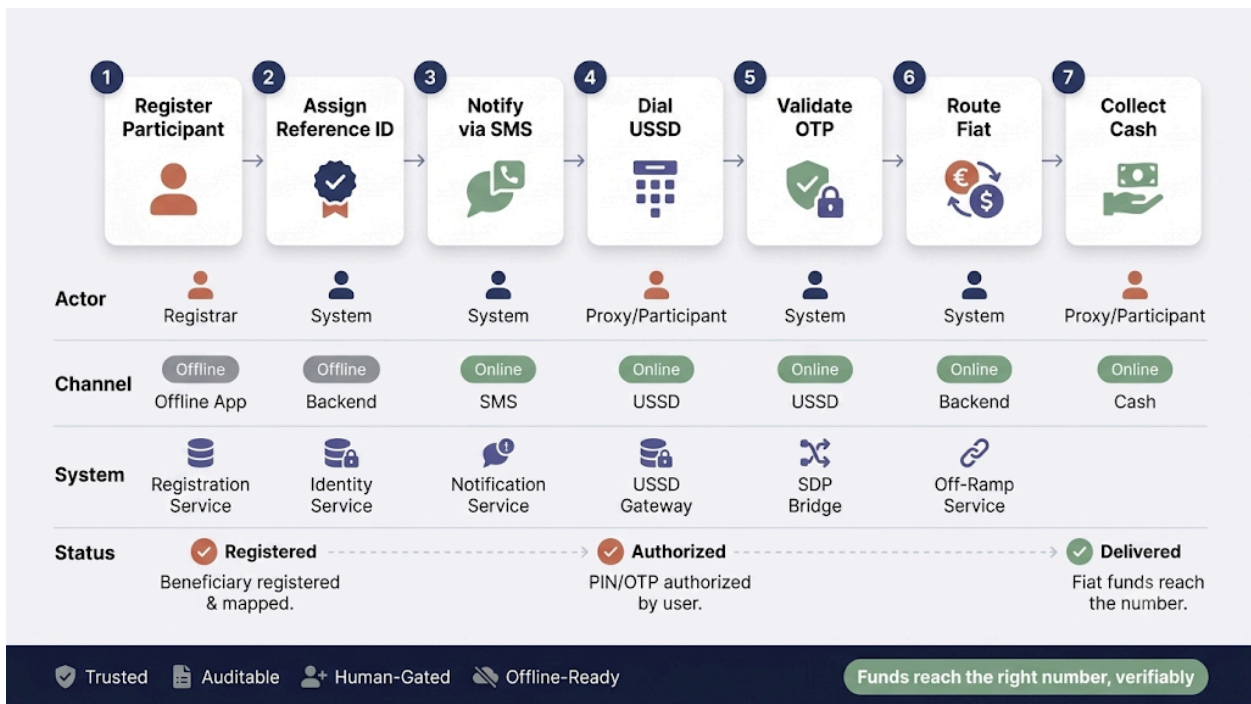


Figure 3 - LastMile delivery use flow, by actor and system component.

Project Schedule

This schedule focuses exclusively on building Product 2: LastMile, SAPCONE's custom offline registration and SMS/USSD-based proxy delivery system. The schedule is divided into two phases. Phase 1 (The Studio Build) covers the current week (Monday, July 13 to Friday, July 17, 2026). Phase 2 (Project Propel) covers the following three months.

Phase 1: Studio Build (This Week)

Goal: Deliver a working end-to-end prototype of the LastMile delivery layer on Stellar Testnet, proving the offline-to-USSD/OTP flow.

Monday, July 13 - The Registrar App & Identity Abstraction

- Task 1 (L-1): Build the Registrar UI (Tablet). Create the unified flow capturing Name, Currency, Number, and the Proxy Checkbox.
- Task 2 (L-1): Implement Reference ID Generation (removing PIN complexity).
- Task 3 (L-3): Enforce PII Isolation. Set up the local database schema so personal data (names) is strictly separated from the transactional identity (Reference ID).

Tuesday, July 14 - Offline Tolerance & Sync Engine

- Task 1 (L-2): Implement the local queueing system. Ensure the Registrar App can store hundreds of new participants locally while in airplane mode.
- Task 2 (L-2): Build the Sync Engine. Create the logic that detects a returning internet connection and securely pushes the queued Reference IDs and metadata to the backend.

Wednesday, July 15 - The Comm Channel & Relayer

- Task 1 (L-4): Set up the SMS Gateway Webhook. Configure the backend to dispatch SDP-generated OTPs to the registered numbers (whether direct or proxy).
- Task 2 (L-4): Build the USSD / SMS Parser. The backend must accurately extract the [Reference ID] and [OTP] from incoming user requests to authenticate the session.
- Task 3 (L-4): Implement Caller ID verification against the proxy database.

Thursday, July 16 - SDP Integration & Admin Oversight

- Task 1 (L-5): Build the SDP OTP verification bridge. Connect the incoming USSD OTP to the Stellar Disbursement Platform to unlock the funds.
- Task 2 (L-6): Develop the Admin Stagnant Funds View. Create a dashboard flagging uncollected disbursements for human review.
- Task 3 (L-7): Build the manual Clawback Authorization button, allowing admins to return stagnant funds to the treasury.

Friday, July 17 - End-to-End Integration & Acceptance

- Task 1: Connect the fiat off-ramp simulator to route funds upon successful OTP validation.
- Task 2: Live Acceptance Walkthrough - register an offline participant (Proxy=True); sync data; receive OTP via SMS; simulate USSD entry of OTP; watch the system trigger the payout; review stagnant funds in the Admin View.

Phase 2: Project Propel (Post-Studio)

Goal: Transition LastMile into a production-ready, highly secure field tool.

Month 1: Infrastructure & The USSD Companion

- USSD Channel Integration: Finalize the live synchronous USSD menu (*340#) with the Telco aggregator for live feature phone access.
- Live Off-Ramp API Integration: Connect the payout trigger to a live mobile money API so users receive fiat (M-Pesa) immediately.

Month 2: Advanced Control & Compliance

- Task 1 (L-8): Automated Reminders & Human Clawbacks. Polish the SMS reminder logic for uncollected funds and harden the Admin clawback execution flow.
- Task 2 (L-9): Right to Erasure. Build the compliance tooling to securely delete a participant's PII from the local identity store.

Month 3: Mainnet Transition & Go-Live

- External Security Review: Conduct a strict audit of the USSD/SMS Relayer vulnerability points.
- Mainnet Deployment: Migrate LastMile's Stellar operations to the public network.
- Go-Live: Execute the first real-value, proxy-led field delivery.

Conclusion

The deployment of the LastMile platform represents a critical evolution in how humanitarian aid is delivered to the most remote and disconnected communities. By addressing the unique realities of pastoralist communities in Turkana County, where network connectivity is sparse and smartphone penetration is low, SAPCONE can bridge the gap between sophisticated digital cash platforms and the last mile of physical delivery.

This solution eliminates the heavy security risks and logistical overhead of manual cash transport, while implementing a robust, trust-based proxy network managed via universal USSD access and automated fiat off-ramping. Most importantly, it maintains strict human-in-the-loop oversight over treasury funds through manual clawback authorizations.

Ultimately, LastMile ensures that SAPCONE can uphold the dignity and privacy of its participants while offering unprecedented transparency to donor organizations. By leveraging feature phones, offline sync capabilities, and trust-based proxy models, SAPCONE will set a new standard for inclusive, accountable, and scalable humanitarian cash assistance.